

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · CHEMISTRY

CHEMISTRY 5070/2
PAPER 2 November 2023
2 hours

Candidate Name: _____

Centre Number: _____ **Candidate Number:** _____

TIME ALLOCATION & INSTRUCTIONS TO CANDIDATES

A copy of the Periodic Table is provided. Answer all questions.

Show all chemical equations with state symbols and numerical working with units.

SECTION A

1. 5.6 g of iron reacts completely with dilute sulfuric acid: $\text{Fe(s)} + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{H}_2(\text{g})$ [Ar: Fe=56, H=1, S=32, O=16. Molar volume of gas at r.t.p. = 24 dm³] (a) Calculate the number of moles of iron reacted. (b) Determine the volume of hydrogen gas evolved at r.t.p. [Stoichiometry & Mole Concept] [6]
- (a) Calculate the moles of iron in 5.6 g. [2]
- (b) Determine the volume of H₂ gas produced at r.t.p. [2]
- (c) State the chemical test and positive observation for hydrogen gas. [2]